

Assessment of Telisotuzumab Vedotin Drug–Drug Interaction Potential Using Physiologically-Based Pharmacokinetic Modeling and Simulations

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Abstract

Telisotuzumab vedotin (Teliso-V; ABBV-399) is an antibody-drug conjugate (ADC) comprised of the c-Met targeting antibody telisotuzumab (ABT-700) conjugated to the potent cytotoxic monomethyl auristatin E (MMAE). Teliso-V has been evaluated for treating solid tumors and is approved for adults with locally-advanced or metastatic non-squamous non-small cell lung cancer with high c-Met protein overexpression ($\geq 50\%$ tumor cells with strong [+3] staining; determined by FDA-approved test), who have received prior systemic therapy. Here, physiologically-based pharmacokinetic (PBPK) modeling was utilized to evaluate Teliso-V drug-drug interaction (DDI) potential. A published PBPK-model for MMAE as the primary metabolite and a newly-developed telisotuzumab model from existing pre-clinical and clinical trial data were used to create a novel Teliso-V PBPK-model. Unconjugated MMAE release was modeled with drug-to-antibody ratio-specific deconjugation rates, with non-specific and catabolic clearance added to capture half-life and overall PK profile. The Teliso-V model was calibrated and validated using observed clinical trial data, requiring dose-normalized exposure %PEs $\leq 50\%$. CYP3A-mediated DDI simulations demonstrated that when Teliso-V was modeled as the victim, a 43% increase and 70% decrease in MMAE AUC were predicted with ketoconazole (strong CYP3A4-inhibitor) and rifampin (strong CYP3A4-inducer) coadministration, respectively. DDI magnitude was comparable to that observed between another approved ADC with the same MMAE payload (brentuximab vedotin) and ketoconazole and rifampin. The current PBPK simulations demonstrated a lack of perpetrator effect of Teliso-V on midazolam, a sensitive CYP3A substrate. The current analysis provides important information on Teliso-V DDI potential and further demonstrates the utility of PBPK models, particularly in oncology, where dedicated DDI studies are challenging.

Keywords

drug–drug interactions, model informed drug development, modeling and simulation, oncology, PBPK, pharmacology

Introduction

Telisotuzumab vedotin (Teliso-V, ABBV-399) is an antibody-drug conjugate (ADC) comprised of the c-Met targeting antibody telisotuzumab (ABT-700) conjugated to the potent cytotoxic monomethyl auristatin E (MMAE) through a vc-linker. Following intravenous administration, Teliso-V binds to c-Met on the surface of tumor cells, is internalized, and then releases the MMAE payload, leading to microtubule function inhibition, critical cellular process disruption, and cell death. Teliso-V specifically targets a patient population with overexpression of c-Met protein in both amplified and non-amplified MET genetic settings. Given that the efficacy of Teliso-V is associated with c-Met expression, it has been developed as a monotherapy to treat epidermal growth factor wild-type locally advanced or metastatic non-squamous non-small cell lung cancer in adults previously treated with systemic therapy and whose tumors have high c-Met overexpression ($\geq 50\%$ tumor cells with strong [+3] staining; determined by FDA-approved test).¹ Recommended Teliso-V dosing

is 1.9 mg/kg (maximum of 190 mg) every 2 weeks (Q2W). The human PK of Teliso-V has been previously characterized in a first-in-human Phase 1/1b study in cancer patients with advanced solid tumors. This included a dose escalation for every 3 weeks (Q3W, 0.15–3.3 mg/kg)² and Q2W (1.6–2.2 mg/kg) dosing,³ as well as a dose expansion phase (Q3W: 2.7 mg/kg, Q2W: 1.9 mg/kg).³ As previously reported, Teliso-V conjugate and MMAE exhibit dose-proportional PK

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across the 1.6–2.2 mg/kg Q2W and 1.2–3.3 mg/kg Q3W dose ranges. The harmonic mean elimination half-life of conjugate and MMAE is ≈ 3 days and ≈ 4 days, respectively.¹

Due to difficulties of conducting dedicated drug–drug interaction (DDI) studies in cancer patients, physiologically-based pharmacokinetic (PBPK) modeling is used in lieu of such studies to understand potential DDIs, particularly in the case of ADCs with known small molecule payloads for which the potential for DDIs has been previously characterized. Such an approach has been used previously for other MMAE-based ADCs, including polatuzumab vedotin⁴ and enfortumab vedotin.⁵ A PBPK model for brentuximab vedotin has also been developed, but was based on data from a DDI study.^{6,7} Similar to Teliso-V, these ADCs have the same MMAE payload attached to a different proprietary antibody using enzyme-cleavable linkers.⁸ Herein, we present a PBPK model developed for Teliso-V. The objectives were to develop and validate a PBPK model to evaluate the DDI potential for Teliso-V. This model was then used to predict the effect of strong CYP3A4 inhibitors or inducers on MMAE exposures following Teliso-V administration and the effect of MMAE on sensitive CYP3A4 substrate exposures. Model development followed both FDA⁹ and EMA¹⁰ guidances and was done by first developing a PBPK model for the naked antibody using PK data from a prior telisotuzumab in vivo study.¹¹ This naked antibody model was then combined with a published MMAE payload model for another ADC (brentuximab vedotin)⁶ to develop a preliminary Teliso-V PBPK model. Separate clinical PK data sets from the phase 1 Teliso-V trial^{2,3} were then used to calibrate and validate our novel Teliso-V model. Upon calibration and validation, the final PBPK model was used to simulate and characterize the impact of CYP3A4 modulators on the exposures of unconjugated MMAE as well as the impact of Teliso-V on sensitive CYP3A4 substrates. The PBPK modeling and simulations described herein predict the clinical DDI potential for Teliso-V, eliminating the need for a dedicated DDI study.

Methods

The current analysis utilized existing clinical trial data for PBPK model calibration and verification (Table 1). The Independent Ethics Committee or Institutional Review Board at each study site approved study protocols, informed consent forms, and recruitment materials before patient enrollment. All studies were registered at ClinicalTrials.gov and were conducted in accordance with the International Conference for Harmonisation guidelines, applicable regulations, and the Declaration

of Helsinki. All patients provided written informed consent before screening.

The Simcyp simulator (V21.0.223.0 [Simcyp V21]; Sheffield, UK) was used for this analysis. PBPK model development was carried out in the following steps:

- Step 1: A PBPK model was developed for telisotuzumab, the naked antibody (ABT-700). Modeling was based on in vivo PK data from a separate study evaluating ABT-700 in patients with solid tumors¹¹ and in vitro antibody-specific parameters (internal proprietary data). The model was calibrated and verified using existing in vivo data for telisotuzumab.¹¹
- Step 2: An existing PBPK model for unconjugated MMAE was utilized.
- Step 3: A linked PBPK model was developed for the ADC, Teliso-V, utilizing the models in Steps 1 and 2 for telisotuzumab and MMAE, respectively. Teliso-V was modeled as the parent compound; unconjugated MMAE as a metabolite. This preliminary Teliso-V PBPK model was calibrated and verified using existing human in vivo data for the conjugated antibody (Teliso-V), total antibody (Teliso-V + naked telisotuzumab), and MMAE payload from Phase 1 Teliso-V studies in cancer patients.^{2,3}

A detailed diagram illustrating model development, calibration, and verification of the Teliso-V PBPK model is shown in Figure 1.

Modeled Teliso-V PK parameters

The “minimal PBPK model” option in Simcyp V21 was utilized for Teliso-V PBPK model simulations, which included a separate liver compartment and a pooled central compartment that represented the systemic circulation. Each tissue compartment was subdivided into vascular, interstitial, intracellular, and endosomal sub-compartments. The overall structure of the tissue subdivision for large molecules in Simcyp V21 is illustrated in Figure 2a.

Large molecules, once infused intravenously, can be passively exchanged between the vascular and interstitial spaces of each tissue; this is generally described using the well-known two-pore model,¹² and is governed by the reflection coefficient (Φ) and diffusion (PS). Large molecules are also taken up by cellular endosomes at the rate, K_{up} . Once inside the endosome, large molecules may undergo neonatal fragment crystallizable receptor (FcR_N)-mediated recycling (governed by binding to FcR_N) at the rate, K_{rc} . An influx of H⁺ ions into the endosome results in an acidic environment that facilitates the interaction between the antibody component of an ADC and FcR_N. Of note, FcR_N expression occurs primarily within the endosomes of endothelial cells.¹³ The FcR_N-bound ADCs are transported outside

Table 1. Summary of Model Simulation Scenarios and Clinical Data Used for Comparison

Study	Drug	Teliso-V Dose(s)	Study Dose Regimen	Simulation Duration	Study Description	Simulation Population	Purpose
Multiple ascending doses (NCT02099058 ^{2,3})	Teliso-V	1.6, 1.9, and 2.2 mg/kg	30 min IV inf Q2W dosing Fasting condition	6 cycles each cycle 2 weeks	Phase I study in subjects with cancer	100 virtual healthy individuals	Model calibration
Multiple ascending doses (NCT02099058 ^{2,3})	Teliso-V	2.4 and 2.7 mg/kg	30 min IV inf Q3W dosing Fasting condition	6 cycles each cycle 3 weeks	Phase I study in subjects with cancer	20-50 years, 50% female 100 virtual healthy individuals	Model verification
Multiple ascending doses (NCT01472016 ¹¹)	Telisotuzumab	5, 10, and 15 mg/kg	30 min IV inf Q3W dosing Fasting condition	6 cycles each cycle 3 weeks	Phase I study in subjects with cancer	20-50 years, 50% female 100 virtual healthy individuals	Model calibration
Rifampin DDI (simulated)	Teliso-V + Rifampin	1.9 mg/kg	Teliso-V: 30 min IV inf on Day 1 of three 14-day cycles Rifampin: 600 mg PO QD from Cycle 1, Day 14 through Cycle 3, Day 14 (29 days)	42 days	Current PBPK model simulation	20-50 years, 50% female 100 virtual healthy individuals	Model application
Ketoconazole DDI (simulated)	Teliso-V + Ketoconazole	1.9 mg/kg	Teliso-V: 30 min IV inf on Day 1 of three 14-day cycles Ketoconazole: 400 mg PO QD from Cycle 2, Day 5 through Cycle 3, Day 14 (24 days)	42 days	Current PBPK model simulation	100 virtual healthy individuals 20-50 years, 50% female	Model application
Midazolam DDI (simulated)	Teliso-V + Midazolam	1.9 mg/kg	Teliso-V: 30 min IV inf on Day 4 of a single cycle Midazolam: 1 mg IV bolus on Days 1 and 6 of a single cycle	17 days	Current PBPK model simulation	100 virtual healthy individuals 20-50 years, 50% female	Model application

inf, infusion; IV, intravenous; Q2W, every 2 weeks; Q3W, every 3 weeks; Teliso-V, telisotuzumab vedotin.

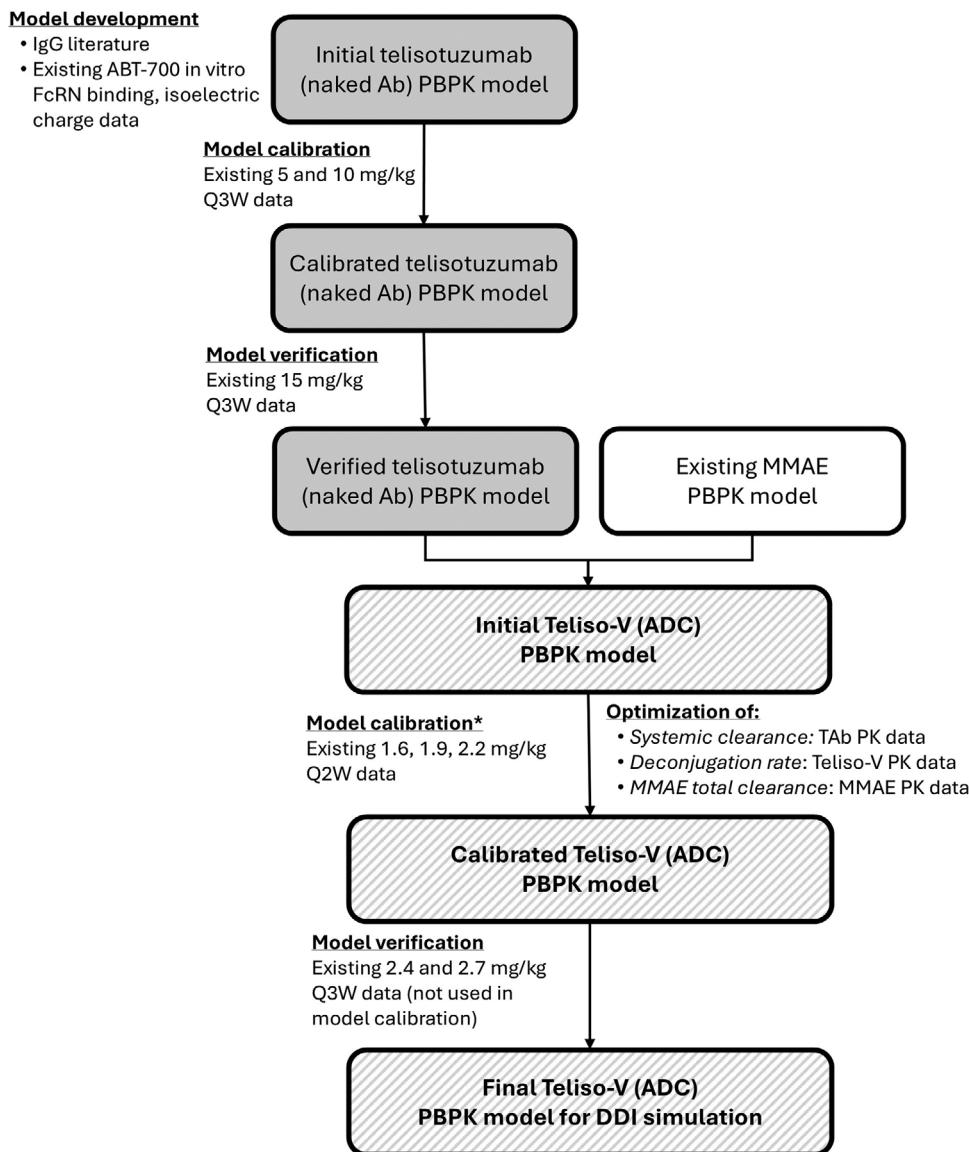


Figure 1. Flowchart of the Teliso-V PBPK model development, calibration, and verification. Ab, antibody; ADC, antibody-drug conjugate; FcRN, neonatal fragment crystallizable receptor; MMAE, monomethyl auristatin E; PBPK, physiologically-based pharmacokinetic; Q2W, every 2 weeks; Q3W, every 3 weeks; TAb, total antibody.

the cell, where the higher physiological pH (7.4) enables ADC release from FcR_N (vascular/interstitial compartments). The acidic environment inside endosomes also facilitates catabolic clearance of large molecules, which may lead to small molecule payload release into the vascular space. Large molecules can also traffic into the intracellular space independent of the endosome via passive means (governed by transport from the interstitial space [PS_I] or via active uptake (governed by clearance from the interstitial [CL_I] and intracellular [CL_T] spaces). FcR_N-based modeling for Teliso-V is consistent with prior modeling of other MMAE-payload ADCs (polatuzumab vedotin,⁴ enfortumab vedotin⁵).

Model Development and Validation

Telisotuzumab (ABT-700, Naked Antibody) PBPK Model. The PBPK model for telisotuzumab (ABT-700), the naked antibody, was developed using Simcyp V21. A minimal PBPK model was developed based on parameters pertaining to wild-type IgG in Simcyp V21 and the FcR_N binding and surface charge properties of telisotuzumab (internal data).

The Simcyp V21 simulator was calibrated toward wild-type IgG.¹⁴ Default values were used except for the FcR_N binding affinity (K_d) and isoelectric charge parameters. Sensitivity analyses were then performed using telisotuzumab K_d and K_{up} . K_d was chosen as

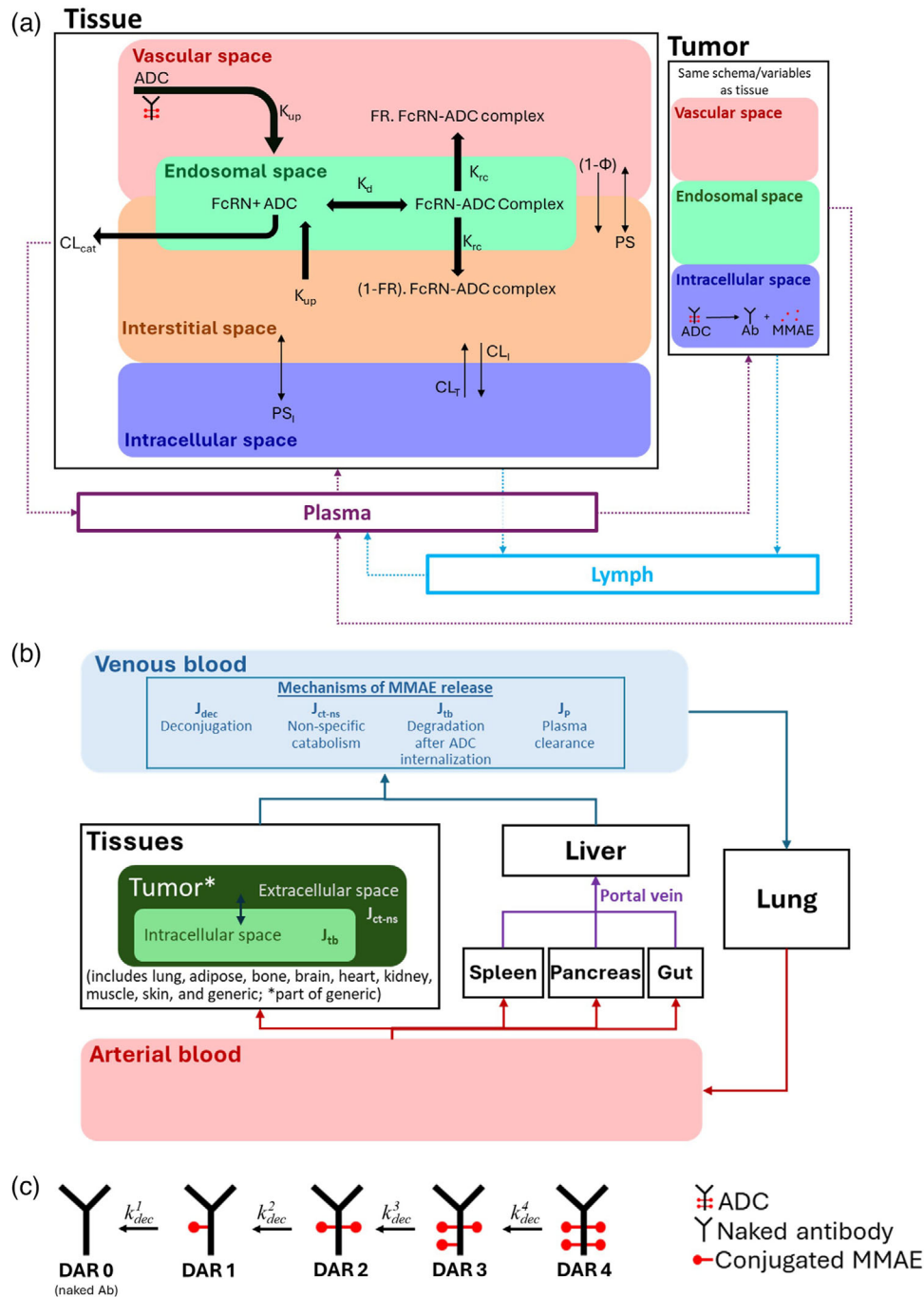


Figure 2. (a) Simplified diagrams of the telisotuzumab vedotin and (b) unconjugated MMAE compartment distribution models utilizing minimal PBPK modeling. (c) Description of MMAE deconjugation from the ADC is also provided. ϕ , interstitial subcompartment movement of antibodies via convective lymph transport; ADC, antibody-drug conjugate; CL_i , clearance from the interstitial space via active pathways; CL_T , clearance from the intracellular space via active pathways; CL_{cat} , clearance via catabolic pathways; DAR, drug-to-antibody ratio; FcRN, neonatal fragment crystallizable receptor; FR, fraction of FcRN-mediated drug complex recycling; J_{xxx} , rate of specified MMAE deconjugation rate; k_{dec}^j , DAR species MMAE deconjugation rate; K_{up} , uptake rate; K_d , FcRN binding affinity; K_{rc} , FcRN drug complex recycling rate; MMAE, monomethyl auristatin E; PS, interstitial subcompartment movement of antibodies via diffusion; PS_i , movement of therapeutic protein (MMAE) to/from the interstitial space via passive pathways.

an input parameter because the telisotuzumab K_d was known independently from that of the wild-type IgG. If the used binding affinity assay led to a different modeled wild-type IgG K_d , then the telisotuzumab K_d would be calibrated. Alternatively, other model parameters (e.g., fraction of FcR_N-mediated drug complex recycling [FR], K_{rc} , K_{up}) could be calibrated. K_{up} was chosen to be optimized because the isoelectric point of telisotuzumab is one unit higher than that of wild-type IgG (8.4 vs 7.4).¹⁵ This may have resulted in electrostatic interaction differences between the plasma membrane and the charged therapeutic protein, leading to a potential difference between telisotuzumab and wildtype IgG endocytosis K_{up} . While both parameters may have provided a comparable fit to clinical data, choosing K_d could warrant further calibration of FR or K_{rc} . Therefore, K_{up} was chosen as the parameter to calibrate in the final model.

Antibodies may also distribute to the interstitial sub-compartment via convective lymph transport and diffusion (PS) through small and large pores in the endothelial wall. The Simcyp V21-predicted values for these parameters were used in the model. Simcyp V21 was used to model therapeutic protein movement into and out of the intracellular compartment via passive (PS_i) and active (CL_T and CL_i) pathways. It should be noted that uptake into the intracellular compartment has not historically been used in this model due to insufficient information.

Naked antibodies are eliminated via target-mediated and/or non-specific cellular uptake and subsequent proteolytic degradation. Non-specific degradation, which involves FcR_N-mediated recycling, predominantly occurs in endothelial cells and mononuclear phagocytes. Target-mediated degradation, which involves both receptor-mediated internalization and intracellular lysosomal degradation, primarily occurs in target-expressing cells.¹⁶ Available Phase 1 clinical trial data at telisotuzumab (naked antibody) doses of 5-25 mg/kg revealed linear PK, with C_{max} proportional to dosing level.¹¹ Because these findings suggested that target-mediated drug disposition (TMDD) did not play a meaningful role, TMDD was not considered in the naked antibody model.

MMAE Payload PBPK Model. The initial unconjugated MMAE PBPK model was built using a previously developed model as described by Chen et al.⁶ and is summarized in Figure 2b. The formation of unconjugated MMAE (implemented as a metabolite of Teliso-V in the full model) is rate-limited by ADC internalization, catabolism, release, and, to a small extent, deconjugation. The clearance of MMAE is known to occur via the CYP3A-mediated metabolic pathway and biliary excretion.⁶ A human mass balance

study following a single dose of brentuximab vedotin showed that the primary route of excretion of unconjugated MMAE was via the feces.⁷ Additionally, in bile-duct cannulated rats, approximately 60% of the total administered unconjugated, radio-labeled MMAE dose was excreted unchanged in bile. These data suggest that approximately 50% of dosed MMAE is cleared as unchanged MMAE via the bile.⁶

In the current analysis, total MMAE clearance was modeled based on the relationship between the clearance of the parent and the metabolite,¹⁷ the above-mentioned literature,⁶ and available clinical data for Teliso-V. Here, Teliso-V is modeled as the parent, as has been done for some ADCs with MMAE payloads.⁵ In support of our approach, a PBPK model for enfortumab vedotin was successfully developed using the ADC as the parent and a similar optimized MMAE clearance (2.72 L/h).⁵

The contribution of P-glycoprotein (P-gp) efflux was not included in the MMAE payload PBPK model. Although MMAE has been shown to be a P-gp substrate in vitro, P-gp mediated clinical DDI is ascribed primarily to inhibition or induction in the gut (not applicable in the case of Teliso-V due to systemic administration and subsequent release of MMAE from the ADC), with limited clinical impact following inhibition or induction in the liver or kidney.^{18,19}

A minimal PBPK model with a single adjusting compartment for MMAE was used in the Chen et al. 2015 model,⁶ but a full PBPK model needed to be used in the current model. Therefore, the volume of distribution needed to be calibrated. To capture the clinical MMAE exposure from Teliso-V administration, the clearance parameter required further calibration. The Simcyp V21 retrograde calculator was used to capture 50% CYP3A4 and 50% biliary elimination when calibration clearance parameters from the base model were applied. In the current analysis, all parameters (except MMAE volume of distribution and clearance) were kept consistent with the Chen et al. 2015 model.⁶

Teliso-V PBPK Model. The link between the Teliso-V PBPK model and the unconjugated MMAE model is the release of MMAE from Teliso-V, the vast majority of which occurs intracellularly following ADC internalization and subsequent catabolism (Figure 2a). This process is governed by deconjugation and catabolism, both of which are modeled in Simcyp V21. For ADCs, elimination that occurs due to antibody-payload deconjugation includes cytotoxic MMAE payload release and subsequent naked antibody (telisotuzumab) elimination via enzymatic and/or chemical processes. This dual pathway may lead to higher ADC than naked antibody elimination. At the time of full Teliso-V model development, clinical data were suggestive of

non-specific systemic clearance. This differed from the telisotuzumab naked antibody PBPK model (also based on clinical data), but non-specific systemic clearance can still be used as a surrogate for antibody uptake into non-FcR_N expressing cells (e.g., hepatocytes) and consequent degradation.¹⁶ Clinical data showing a longer half-life of telisotuzumab (naked antibody; 14.1 days¹¹) than Teliso-V (ADC, approximately 3 days¹) supports such degradation. By matching the observed data and half-life of both total antibody (TAb) and conjugated antibody, the non-specific systemic plasma clearance was determined. Therefore, the best fit of the clinically observed data accounted for this additional clearance. Available clinical data at the clinically-relevant dose range of Teliso-V (ADC; 1.6-3.0 mg/kg) do not suggest target-mediated drug disposition (TMDD). As a result, TMDD was not considered in our model.

The initial value of the deconjugation rate for Teliso-V was calculated using data from an in-house in vitro study, with the 24-h MMAE concentration change estimated as the initial deconjugation rate. This was previously calculated using an MMAE spike into plasma at the low, mid, and high quality control levels with an initial Teliso-V spike of 0.476 mg/mL (internal AbbVie study). Of note, this available deconjugation rate was used in initial model development but was later calibrated to human in vivo data (Teliso-V PK profile) using Simcyp V21. The deconjugation rate for the drug-to-antibody ratio (DAR) 1 value was calibrated, and the Simcyp calculator was used to determine the deconjugation rate for higher DAR species. Within Simcyp, deconjugation in plasma is modeled by transit compartments with assumed first-order rates for the transition from a DAR species to its neighboring lower DAR species (Figure 2c), accompanied by drug release. The following equation was used in Simcyp (Equation (1)):

$$k_{\text{dec}}^j = k_{\text{dec}}^j [1 + (j - 1)\theta]; j = 0, 1, 2, 3, 4 \quad (1)$$

where, k_{dec}^j is the deconjugation rate and $\theta = 0.315$ (Simcyp default). The k_{dec}^j was optimized for DAR species 1 ($j = 1$); the remaining deconjugation rates were automatically calculated in Simcyp. A sensitivity analysis was performed using the DAR 1 species deconjugation rate constant in Simcyp V21. Ultimately, a deconjugation rate value was chosen so that the conjugated antibody C_{max} , AUC, and $t_{1/2}$ best matched that of the clinical data. The key assumptions of the PBPK model were as follows:

1. Simcyp V21 calibrated IgG parameters (K_{rc} , FR) were used even though the Teliso-V binding affin-

ity to FcR_N was measured without a wild-type comparison.

2. Teliso-V was assumed to have no TMDD (based on clinical data in the dose range examined [1.6-3.0 mg/kg]).
3. All released MMAE payload via Teliso-V catabolism was assumed to have instantaneous release into the vascular space.
4. Teliso-V deconjugation was assumed to occur at the same rate in the plasma and the tissues.

Final model parameters, along with the model calibration and verification data sources, are fully described in the Supporting Information.

Simulation results in a cancer population and a healthy population did not meaningfully differ (Table S1). Therefore, PBPK model simulations were carried out within Simcyp V21 using the Sim-Healthy volunteers virtual population. Based on the typical healthy volunteer profile, virtual patients were 20-50 years of age, and 50% were female. Study simulation details are presented in Table 1. Data from Teliso-V studies were used for model calibration and verification, with simulations based on study replication. For DDI simulations with perpetrator co-medications (ketoconazole, rifampin, midazolam), the existing library models within Simcyp V21 were used without any changes.

The ability of the PBPK model to predict observed in vivo data from Phase 1 pharmacokinetic studies was evaluated and quantified based on the percent model prediction error (%PE, Equation (2)). Predefined rules for acceptable %PE were set, and the model was adjusted until the %PE was $\leq 50\%$.

$$\%PE = \frac{(\text{observed value} - \text{predicted value})}{\text{observed value}} \times 100 \quad (2)$$

Because in vivo data can be highly variable, a higher model prediction error can be acceptable. For example, PK parameters with a prediction error of 90% would be acceptable for in vivo data with a coefficient of variation (CV) of 40%.²⁰ A stricter acceptance criteria of 50% was used for the current model predictions despite the high variability of observed in vivo data for Teliso-V (up to 95% CV) and MMAE PK parameters for clinically relevant dose levels. Though model simulations were done on a virtual healthy volunteer population, in vivo PK data from cancer patients (which can have even higher variability) were used for model calibration and verification. Therefore, a 50% acceptance limit for PE was deemed appropriate and would have enough precision to make meaningful clinical recommendations.

Table 2. Calibrated Teliso-V PBPK Model-Predicted Versus Observed PK Parameters Following Q3W Monotherapy Dosing

Dose		PK Parameter	Predicted ^a	Observed ^a	%PE
Conjugated antibody					
2.4 mg/kg	(N = 5)	C _{max} (μg/mL)	55.1	48.8 (29)	13
		AUC _{tau} (μg•h/mL)	3520	3800 (40)	−7
		t _{1/2} (d) ^b	2.10	3.15 (0.80)	−33
2.7 mg/kg	(N = 12)	C _{max} (μg/mL)	62.0	43.6 (22)	42
		AUC _{tau} (μg•h/mL)	3960	3520 (28) ^d	13
		t _{1/2} (d) ^b	2.10	3.44 (1.60) ^d	−39
Total antibody					
2.4 mg/kg	(N = 5)	C _{max} (μg/mL)	55.1	72.8 (47)	−24
		AUC _{tau} (μg•h/mL)	4100	7710 (48)	−47
		t _{1/2} (d) ^b	3.37	3.68 (1.12)	−8
2.7 mg/kg	(N = 12)	C _{max} (μg/mL)	62.0	50.2 (23)	24
		AUC _{tau} (μg•h/mL)	4610	6470 (31) ^d	−29
		t _{1/2} (d) ^b	3.37	3.53 (1.69) ^d	−5
MMAE					
2.4 mg/kg	(N = 6)	T _{max} (h) ^c	112	85.0 (42.2, 168.2)	32
		C _{max} (ng/mL)	2.65	3.04 (45)	−13
		AUC _{tau} (ng•h/mL)	778	779 (48) ^e	0
2.7 mg/kg	(N = 12)	T _{max} (h) ^c	113	104.4 (39.7, 172.8)	8
		C _{max} (ng/mL)	3.00	3.94 (42)	−24
		AUC _{tau} (ng•h/mL)	884	861 (39) ^f	3

%PE, model prediction error; ADC, antibody drug conjugate; AUC_{tau}, area under the curve over dosing cycle (3 weeks); C_{max}, maximum concentration; MMAE, monomethyl auristatin E; Q3W, every 3 weeks; T_{max}, time of maximum concentration; t_{1/2}, half-life.

^aObserved and predicted values are geometric means of the individual results.

^bPresented as harmonic mean (pseudo SD).

^cPresented as median (min, max).

^dN = 11.

^eN = 5.

^fN = 7.

Model Calibration and Verification

The Teliso-V PBPK model was developed using a combined bottom-up and top-down modeling approach. The base model for telisotuzumab (naked antibody) was developed using IgG parameters¹⁴ and in vitro data for the telisotuzumab.¹¹ Observed PK parameters (Cycle 1 of Q3W monotherapy dosing) from clinical trial data¹¹ were used to calibrate and validate the model (Tables S2 and S3). K_{up} was the only parameter calibrated with clinical data during model calibration.

The base model developed for telisotuzumab (ABT-700) did not adequately represent PK data for TAB, conjugated antibody (Teliso-V), or unconjugated MMAE. Hence, a second calibration step was needed using human in vivo data from different analytes as follows:

- Step 1: Non-specific plasma clearance was added with a catabolic clearance. This allowed the capture of the TAB (Teliso-V + telisotuzumab) PK profile, t_{1/2}, and in vitro data-based deconjugation rate.
- Step 2: The deconjugation rate was calibrated to capture the Teliso-V (ADC) PK data. This was justified because in vitro data-estimated MMAE deconjugation rate did not describe both the MMAE and Teliso-V exposure well.

- Step 3: Optimization of MMAE distribution volume and clearance. This was done by calibrating the TAB and Teliso-V models to human in vivo data of the MMAE payload exposure from Teliso-V deconjugation.

The final PBPK model for Teliso-V was verified using human in vivo PK data (2.4 and 2.7 mg/kg Q3W), as summarized in Table 2. Of note, these data were exclusively used for model verification; a different data set was used for model calibration.

A sensitivity analysis was performed for the PBPK model-derived parameters, including Teliso-V deconjugation rate and total MMAE clearance (Figure S1 and Table S4). These analyses were of great importance because the deconjugation rate meaningfully impacts both MMAE formation and the Teliso-V degradation. Sensitivity analyses also examined the potential impact of deconjugation rate and non-specific systemic clearance on Teliso-V DDIs (Table S5).

Model Application for Prediction of Drug-Drug Interactions

Simulated “trials” for model DDI predictions (Figure 3) were based on historical practices for ADC DDI



Figure 3. Schematic of “trial” simulations to examine potential drug–drug interactions when Teliso-V was coadministered with (a) ketoconazole (CYP3A4 inhibitor; Teliso-V as victim, simulation duration: 1008 h [Days 1–43; 42 days]), (b) rifampin (strong CYP3A4 inducer; Teliso-V modeled as victim, simulation duration: 1008 h [Days 1–43; 42 days]), and (c) midazolam (CYP3A4 competitive inhibitor; Teliso-V modeled as perpetrator; simulation duration: 408 h [Days 1–18; 17 days]). All simulations started on Cycle 1, Day 1 at “9:00 a.m.”.

simulations and human in vivo DDI studies conducted for brentuximab vedotin.⁶

Teliso-V as Victim. Teliso-V has not been previously evaluated as the victim of enzymatic DDI in in vivo studies. However, brentuximab vedotin, another approved MMAE-based ADC, was evaluated in PK studies for interactions with ketoconazole (strong CYP3A4 inhibitor) and rifampin (strong CYP3A4 inducer).⁷ Those findings were used for Teliso-V model comparison. Simulations were conducted using the developed PBPK models for Teliso-V with ketoconazole and rifampin, replicating the brentuximab vedotin PK study as closely as possible. Of note, the Teliso-V model had a 14-day cycle (1.9 mg/kg Q2W [based on the recommended phase 2 dose]), but the brentuximab vedotin model had a 21-day cycle based on the recommended Q3W dosing regimen.⁷ Simulation scenarios and associated clinical data used for comparison are fully described in Table 1. Briefly, the simulations were conducted over three 14-day cycles. In Cycle 1, Teliso-V was dosed alone; in Cycles 2 and 3, Teliso-V was dosed with a perpetrator (ketoconazole [400 mg/day orally; Cycle 2, Day 5 through Cycle 3, Day 14] or rifampin [600 mg/day orally; Cycle 1, Day 14 through Cycle 3,

Day 14]). DDI ratios were estimated in Cycle 3 to ensure maximum perpetrator impact on Teliso-V PK.

Teliso-V as Perpetrator. Prior in vitro assessment demonstrated that MMAE is a competitive inhibitor of CYP3A4 (half maximal inhibitory concentration [IC₅₀]: 10 μM) and a quasi-irreversible metabolism-dependent inhibitor of CYP3A4 (maximum rate of inactivation [k_{inact}]: 0.10 min^{−1}, inhibitory constant [K_i]: 1.12 μM).⁶ However, DDI calculations on in vivo data suggest that MMAE is a substrate of CYP3A but not an inhibitor or inducer of CYP3A.⁷ The developed PBPK model for Teliso-V (and the MMAE payload) was simulated using a Q2W Teliso-V dose of 1.9 mg/kg administered with midazolam (1 mg IV bolus; Days 1 and 6), a sensitive CYP3A4 substrate. Modeling was performed over a single cycle from Day 1 to Day 18 (17-day duration). The influence of varying inhibition parameters is provided in Table S6.

Results

Teliso-V PBPK Model Performance

The calibrated PBPK model performance was assessed using in vivo trial data obtained with Q3W dosing of 2.4 and 2.7 mg/kg Teliso-V (3.0 mg/kg examined,

Table 3. Teliso-V PBPK Model Predicted Versus Observed Parameters Following Q2W Monotherapy Dosing

Dose		PK Parameter	Predicted ^a	Observed ^a	%PE
Teliso-V (ADC)					
1.6 mg/kg	(N = 10)	C _{max} (μg/mL)	36.7	36.7 (31)	0
		AUC _{tau} (μg•h/mL)	2310	2040 (36)	13
		t _{1/2} (d) ^b	2.25	2.43 (0.88)	-7
1.9 mg/kg	(N = 7)	C _{max} (μg/mL)	43.6	29.0 (43)	50
		AUC _{tau} (μg•h/mL)	2740	2130 (55) ^d	29
		t _{1/2} (d) ^b	2.25	3.02 (0.83) ^d	-25
2.2 mg/kg	(N = 3)	C _{max} (μg/mL)	50.5	40.4 (12)	25
		AUC _{tau} (μg•h/mL)	3170	2530 (9)	25
		t _{1/2} (d) ^b	2.25	3.13 (0.70)	-28
Total antibody					
1.6 mg/kg	(N = 10)	C _{max} (μg/mL)	36.7	50.3 (68)	-27
		AUC _{tau} (μg•h/mL)	2610	4100 (41)	-36
		t _{1/2} (d) ^b	3.37	2.60 (0.94)	30
1.9 mg/kg	(N = 7)	C _{max} (μg/mL)	43.6	33.4 (26)	31
		AUC _{tau} (μg•h/mL)	3090	3780 (50) ^d	-18
		t _{1/2} (d) ^b	3.37	3.36 (0.92) ^d	0
2.2 mg/kg	(N = 3)	C _{max} (μg/mL)	50.5	62.4 (71)	-19
		AUC _{tau} (μg•h/mL)	3580	4640 (9)	-23
		t _{1/2} (d) ^b	3.37	4.13 (0.85)	-18
MMAE payload					
1.6 mg/kg	(N = 10)	T _{max} (h) ^c	109	80.1 (28.0, 168.6)	36
		C _{max} (ng/mL)	1.75	1.68 (46)	4
		AUC _{tau} (ng•h/mL)	426	424 (45) ^d	0
1.9 mg/kg	(N = 6)	T _{max} (h) ^c	110	117.1 (71.0, 173.4)	-6
		C _{max} (ng/mL)	2.09	2.24 (53)	-7
		AUC _{tau} (ng•h/mL)	511	405 (64) ^e	26
2.2 mg/kg	(N = 3)	T _{max} (h) ^c	111	137.6 (71.1, 166.4)	-20
		C _{max} (ng/mL)	2.42	2.95 (95)	-18
		AUC _{tau} (ng•h/mL)	596	620 ^f	-4

%PE, model prediction error; ADC, antibody drug conjugate; AUC_{tau}, area under the curve over dosing cycle (2 weeks); C_{max}, maximum concentration; MMAE, monomethyl auristatin E; Q2W, every 2 weeks; t_{1/2}, half-life; T_{max}, time of maximum concentration.

^aObserved and predicted values are geometric means of the individual results.

^bPresented as harmonic mean (pseudo SD).

^cPresented as median (min, max).

^dN = 6.

^eN = 4.

^fN = 1.

but the dosing sample size was too small to include [N = 2]). Of note, model development was done with a different Q2W dosing data set (1.6, 1.9, and 2.2 mg/kg). Table 3 summarizes the observed and model-predicted PK parameters for TAB, conjugated antibody, and MMAE. All prediction errors were <50% for TAB and conjugated antibody, indicating adequate model performance. Dose-normalized PBPK model predictions of analyte concentrations over time closely followed dose-normalized pooled concentrations of in vivo trial data (Figure 4). Model predictions and observed data that were not dose-normalized are shown in the Supporting Information (Figure S2). The prediction errors were within 50% for the Teliso-V (conjugated antibody) and TAB and within 36% for the MMAE payload (Table 3).

The dose-normalized final PBPK model-derived exposures of telisotuzumab (naked antibody), MMAE, and Teliso-V (ADC; C_{max} and AUC) were compa-

table to pooled, dose-normalized observed exposures (Figure 4).² TAB, Teliso-V, and unconjugated MMAE PK parameters derived from the initial PBPK model were also similar to those derived from in vivo data for Q2W dosing (all %PE ≤ 50%; Table 3 and Figure S3). The T_{max} for TAB and Teliso-V was underpredicted by the model across all dose ranges owing to the first post-infusion sampling time point in the clinical study. The Simcyp V21 PBPK model predicted that T_{max} in the venous compartment would occur immediately after completion of a 30-min IV infusion, while the first in vivo PK sample was obtained 0.5 h later (30 min after the end of the infusion). As a result, the predicted and observed T_{max} were approximately 0.5 and 1 h, respectively, for both TAB and Teliso-V at most Teliso-V doses.

The initial PBPK model (prior to calibration) adequately captured dose-normalized concentrations of the MMAE payload, with a %PE ≤ 50 for C_{max} and

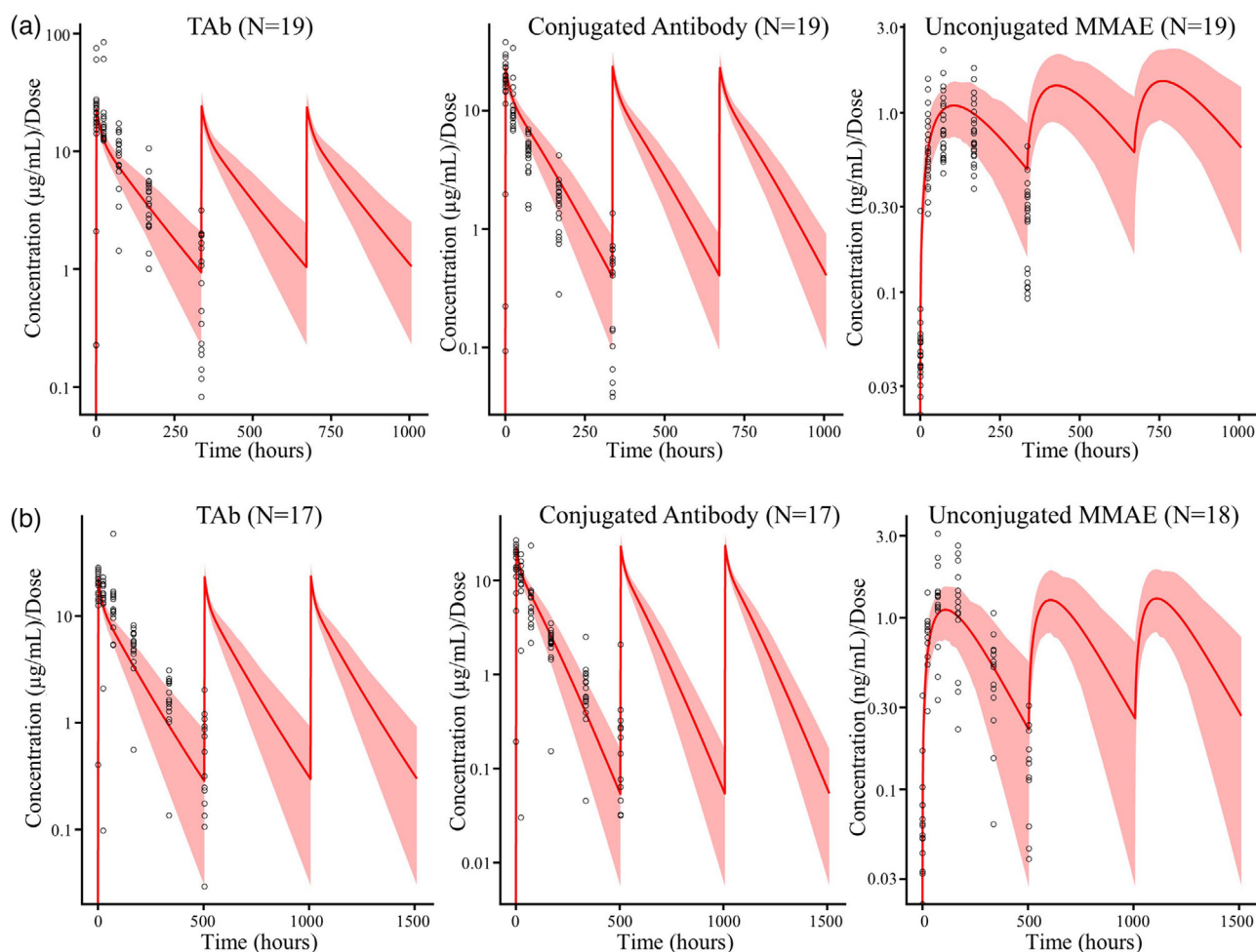


Figure 4. (a) PBPK model calibration and (b) verification for TAb, conjugated antibody (telisotuzumab vedotin), and unconjugated MMAE payload concentrations. (a) Dose-normalized mean concentration profile of the initial model and existing clinical trial data (1.6, 1.9, and 2.2 mg/kg every 2 weeks via IV infusion). PK parameters were optimized to obtain the best fit of predicted versus observed concentrations. (b) Dose-normalized mean concentration profiles of the calibrated PBPK model and existing clinical trial data (2.4 and 2.7 mg/kg every 3 weeks via IV infusion). Open circles represent patient-level clinical trial data (NCT02099058). The solid line represents the mean model prediction. Shaded regions represent the 5th and 95th percentiles. MMAE, monomethyl auristatin E; TAb, total antibody.

AUC_{tau} . However, predicted and observed MMAE plasma concentrations did not always match the clinical data at each dose level respectively, particularly for 2.2 mg/kg Q2W dosing (Figure S3). As such, the sample size for this dosing was small ($N = 3$ patients) and the data were highly variable.

PBPK Model Predictions of Teliso-V CYP3A-Mediated Drug–Drug Interactions

Teliso-V as Victim. Potential DDIs of unconjugated MMAE due to CYP3A4 inhibition or induction were evaluated using the validated PBPK model and are summarized in Table 4. Simulation parameters are detailed in Table 1. Teliso-V (1.9 mg/kg IV infusion Q2W [Day 1, Cycles 1-3]) with ketoconazole

coadministration (oral 400 mg/day; Cycle 2, Day 5 through Cycle 3, Day 14) was predicted to result in a 30% increase in C_{max} (geometric mean ratio [GMR, 90%CI]: 1.30 [1.28-1.33]) and a 43% increase in AUC_{tau} (GMR [90%CI]: 1.43 [1.39-1.47]) for unconjugated MMAE. Model predictions with rifampin coadministration (oral 600 mg/day; Cycle 1, Day 14 through Cycle 3, Day 14) indicated a 56% decrease in C_{max} (GMR [90%CI]: 0.44 [0.42-0.46]) and a 70% decrease in AUC_{tau} (GMR [90%CI]: 0.30 [0.28-0.32]) for unconjugated MMAE. These findings were similar to those of other MMAE-based ADCs (Table 4).

Teliso-V as Perpetrator. There was no meaningful effect of Teliso-V (1.9 mg/kg IV infusion on Day 4 of

Table 4. PBPK Model Based DDI Predictions for Teliso-V and Other ADCs

ADC	DDI Ratios ^a	ADC as Victim		ADC as Perpetrator
		Ketoconazole (CYP3A4 Inhibitor)	Rifampin (CYP3A4 Inducer)	Midazolam (CYP3A4 Inhibitor)
Telisotuzumab vedotin (1.9 mg/kg Q2W)	C _{max} ratio	1.30 (1.28-1.33)	0.44 (0.42-0.46)	1.00 (1.00-1.00) ^b
	AUC ratio	1.43 (1.39-1.47)	0.30 (0.28-0.32)	1.06 (1.05-1.06) ^b
Brentuximab vedotin ⁷	C _{max} ratio	1.25	0.56	1.15
	AUC ratio	1.34	0.54	0.94
Enfortumab vedotin ⁵	C _{max} ratio	1.15	0.72	1.00
	AUC ratio	1.38	0.47	1.14
Polatuzumab vedotin ⁴	C _{max} ratio	1.18	0.69	1.00
	AUC ratio	1.48	0.49	1.00

ADC, antibody-drug conjugate; AUC, area under the curve; DDI, drug-drug interaction; Q2W, every 2 weeks; Q3W, every 3 weeks.

Brentuximab vedotin: 1.8 mg/kg Q3W for midazolam and rifampin coadministration, 1.2 mg/kg Q3W for ketoconazole coadministration;⁷ enfortumab vedotin: single 1.25 mg/kg dose;⁵ polatuzumab vedotin: single 1.8 mg/kg dose.⁴

^aGeometric mean (90% confidence interval [where applicable]); AUC_{inf} ratio for brentuximab vedotin, geometric mean AUC_{tau} ratio for Teliso-V (predicted), enfortumab vedotin, and polatuzumab vedotin (tau = cycle length of recommended or clinically meaningful dosing interval), AUC₀₋₇₂ ratio for Teliso-V + midazolam.

^bMidazolam MMAE inhibition parameters for Teliso-V modeling: K_i = 5 μM, K_{app} = 1.12 μM, K_{inact} = 0.10 min⁻¹.

a 14-day cycle) on midazolam (1 mg IV bolus, Days 1 and 6), with the model predicting no change in midazolam C_{max} (GMR [90%CI]: 1.00 [1.00-1.00]) and a 6% increase in AUC₀₋₇₂ (GMR [90%CI]: 1.06 [1.05-1.06]). The predicted GMRs for midazolam (AUC and C_{max}), with and without Teliso-V co-administration are summarized in Table 4.

Discussion

A PBPK model to predict Teliso-V DDIs was developed. A clinical DDI study was conducted for the first MMAE-based ADC, brentuximab vedotin.⁷ Using these in vivo data, Chen et al.⁶ developed and validated a PBPK model. This model was later applied to determine polatuzumab vedotin potential for DDIs, which guided DDI-related labeling without a dedicated DDI clinical study.⁴ Because enfortumab vedotin had the same MMAE payload as these other ADCs, this practice was also adopted for enfortumab vedotin.²¹ Of note, no DDI-dedicated clinical studies were performed for either polatuzumab vedotin²² or enfortumab vedotin,²¹ but product labels describe PBPK model findings and DDI-related predictions.

A mechanistic PBPK model was successfully developed for Teliso-V, an ADC consisting of an anti-cMET monoclonal antibody (ABT-700) linked to an MMAE payload. The model was based on data available from in vitro studies, in vivo studies, and the literature and was developed in a stepwise fashion, considering the naked antibody, the MMAE payload, and the linked ADC in that order. Specific parameters affecting each moiety were separately explored and calibrated using available data. Of importance, the PBPK model was optimized using in vivo data and verified using a

different in vivo data set. Further, comparisons of predicted (PBPK model simulations) and actual (clinical trial data from cancer patients) concentration-time profiles were performed for TAb, conjugated antibody, and unconjugated MMAE for multiple Teliso-V doses (range: 1.6-2.7 mg/kg).

The PBPK model predicted a 43% increase in unconjugated MMAE exposure when Teliso-V was co-administered (as a victim) with ketoconazole, a strong CYP3A4 and P-gp inhibitor. Given that higher unconjugated MMAE exposures are associated with increased rates of Grade ≥3 adverse reactions, these PBPK evaluations informed dosing recommendations that patients should be closely monitored for adverse events when Teliso-V is concomitantly used with a strong CYP3A4 inhibitor.¹ Of note, P-gp mediated clinical DDI is primarily due to inhibition/induction in the gut and Teliso-V is directly administered to the systemic circulation. Further, intracellular enzymes that cleave the antibody-payload vc-linker are frequently overexpressed within tumor cells.^{23,24} As a result, MMAE release from the ADC is largely limited to inside targeted cells. Therefore, Teliso-V has limited clinical impact following P-gp inhibition or induction in the liver or kidney.^{18,19}

Although these Teliso-V DDI predictions have not been verified with an in vivo study, published data from a DDI study with brentuximab vedotin, an ADC with the same MMAE payload, showed a similar MMAE increase (GMR [90%CI] C_{max}: 1.25 [0.90-1.72], AUC_{0-∞}: 1.34 [0.98-1.84]).⁷ It is important to note that dose-normalized unconjugated MMAE exposure from brentuximab vedotin at a single dose of 1.8 mg/kg⁷ is approximately twice that of Teliso-V with Q2W 1.9 mg/kg

dosing. DDIs between ketoconazole and other approved ADCs (enfortumab vedotin,⁵ polatuzumab vedotin [PBPK predictive modeling]⁴) were also consistent with our DDI findings (Table 4). MMAE exposure was predicted to decrease by 70% when Teliso-V was co-administered with a strong CYP3A4 inducer (rifampin). This prediction was also comparable with that reported for other approved ADCs.

The PBPK model predicted minimal-to-no impact of DDIs when Teliso-V was modeled as the perpetrator. Similar to a prior model that evaluated brentuximab vedotin DDIs,⁷ Teliso-V was not predicted to meaningfully change midazolam (a sensitive substrate) exposure (GMR [90%CI]: C_{max} : 1.00 [1.00-1.00], AUC_{0-72} : 1.06 [1.05-1.06]). Thus, Teliso-V was not predicted to inhibit CYP3A4 substrates. This finding is in agreement with prior PBPK model-predicted DDIs of other approved ADCs and CYP3A4 substrates (Table 4). Further, sensitivity analyses suggested an approximate 30% increase in a typical CYP3A4 substrate exposure in the worst-case scenario. Together, the Teliso-V as victim and perpetrator analyses informed dosing recommendations for Teliso-V when co-administered with other medications. Although the payload (MMAE) is the same for Teliso-V and brentuximab vedotin, the DDI between MMAE and a CYP3A4 inhibitor or inducer could differ because of varying MMAE formation rate differences. In fact, discrepancies between Teliso-V and brentuximab vedotin in catabolism and ADC deconjugation have been shown to result in slightly different calibrated clearance values, a required parameter for the MMAE PBPK modelling used here (applied total MMAE clearance for Teliso-V: 2.74 L/h; previously reported total MMAE clearance for brentuximab vedotin: 8 L/h⁶). However, the impact of this difference on the overall predicted DDI liability of Teliso-V was minimal ($\leq 15\%$ increase/decrease). The calibrated value of the total MMAE clearance was similar to a recently published PBPK model-derived value for conjugated MMAE after enfortumab vedotin dosing.⁵ These MMAE formation and elimination rates could have impacted model DDI predictions. However, thorough sensitivity analyses (Table S5) indicated that calibrated model parameters did not meaningfully impact overall Teliso-V DDI liability (up to $\approx 15\%$ increase/decrease).

Conclusion

In conclusion, the developed PBPK model for Teliso-V met the pre-determined acceptance criteria. Simulations using our novel model for Teliso-V exposure provided guidance on predicted DDIs between Teliso-V and CYP3A4 inhibitors/inducers and P-gp inhibitors, eliminated the need for a clinical DDI study, and informed labeling. When Teliso-V was modeled as

victim, unconjugated MMAE exposure (AUC) was predicted to increase by 43% when co-administered with a strong CYP3A4 and P-gp inhibitor. As a result, it is recommended that patients with concomitant use of Teliso-V and a strong CYP3A inhibitor be monitored for adverse events. Conversely, MMAE exposure was predicted to decrease by 70% when the ADC was co-administered with a strong CYP3A4 inducer. When Teliso-V was modeled as the perpetrator and co-administered with a competitive inhibitor of CYP3A4 (midazolam), midazolam exposure was not predicted to meaningfully change. In the absence of a dedicated clinical study, these PBPK model predications of Teliso-V DDIs provided valuable information to guide product labeling and dosing recommendations in the presence of CYP3A4 substrates and perpetrators.

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Author Contributions

Participated in research design: Md Mahbubul Huq Riad, Patrick Marroum, and Apurvasena Parikh. Performed data analysis: Md Mahbubul Huq Riad and Patrick Marroum. Performed data interpretation: Md Mahbubul Huq Riad, Priya Brunsdon, Rajeev Menon, Patrick Marroum, and Apurvasena Parikh. Wrote or contributed to manuscript writing: Md Mahbubul Huq Riad, Priya Brunsdon, Rajeev Menon, Patrick Marroum, and Apurvasena Parikh. All authors approved the final manuscript and agree to be accountable for all aspects of the work.

Conflicts of Interest

All authors are employees of AbbVie and may hold stock or stock options in AbbVie.

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Data Availability Statement

AbbVie is committed to responsible data sharing regarding the clinical trials we sponsor. This includes access to anonymized, individual, and trial-level data (analysis data sets), as well as other information (e.g., protocols, clinical study reports, or analysis plans), as long as the trials are not

part of an ongoing or planned regulatory submission. This includes requests for clinical trial data for unlicensed products and indications. These clinical trial data can be requested by any qualified researchers who engage in rigorous, independent, scientific research, and will be provided following review and approval of a research proposal, Statistical Analysis Plan (SAP), and execution of a Data Sharing Agreement (DSA). Data requests can be submitted at any time after approval in the US and Europe and after acceptance of this manuscript for publication. The data will be accessible for 12 months, with possible extensions considered. For more information on the process or to submit a request, visit the following link: <https://www.abbvieclinicaltrials.com/hcp/data-sharing/.html>.

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Supplemental Information

Additional supplemental information can be found by clicking the Supplements link in the PDF toolbar or the Supplemental Information section at the end of web-based version of this article.